

International Institute of Information Technology, Hyderabad
(Deemed to be University)

Behavioral Research: Statistical Methods (CG.3.403) Spring-2025
Question cum Answer Booklet
EndSem

Max. Time: 120 Minutes

Max. Marks: 45

Roll No: Programme: Date:

Room No: Seat No: Invigilator's Signature:

Special Instructions about the exam

1. This is a closed book exam and all questions are compulsory.
2. There are 13 objective questions and 6 descriptive questions for a total of 45 marks.
3. *In case of any doubt, please answer while listing your assumptions appropriately.*

General Instructions to the students

1. Place your Permanent / Temporary Student ID card on the desk during the examination for verification by the Invigilator.
2. Use of calculators is not allowed. Reading material such as books (unless open book exam) are not allowed inside the examination hall.
3. Borrowing writing material or calculators from other students in the examination hall is prohibited.
4. If any student is found indulging in malpractice or copying in the examination hall, the student will be given 'F' grade for the course and may be debarred from writing other examinations.

Good Luck

12/13

Objective Questions (13 marks):

1. Which of the following best describes a *confound* in the context of experimental research design?

- A) A measurement that cannot be replicated across trials
- B) A variable that the researcher intentionally manipulates
- C) An unmeasured variable that systematically varies with the independent variable and may affect the dependent variable
- D) A random error that occurs during the process of data collection

2. A psychologist is studying the effect of sleep deprivation on problem-solving ability. Participants are randomly assigned to either get 8 hours of sleep or only 4 hours. However, participants in the 4-hour group report feeling stressed because they had to stay in the lab overnight. What is the best way to control for this confound in future studies?

- A) Increase the number of participants in the sleep-deprived group
- B) Allow participants to choose their sleep condition
- C) Ensure both groups spend the night in the lab, regardless of sleep duration
- D) Use a different task to measure problem-solving ability

3. A clinical trial tests whether a new drug lowers blood pressure. The frequentist analysis yields a p-value of 0.03, while the Bayesian analysis produces a Bayes Factor $BF_{10}=1.2$ favoring the alternative hypothesis. What does this difference imply?

- A) Both methods provide strong evidence for the alternative hypothesis
- B) The p-value and Bayes Factor always agree when the sample size is large
- C) The p-value indicates statistical significance, while the Bayes Factor suggests only weak evidence for the alternative
- D) The Bayes Factor invalidates the frequentist result due to a mismatch in priors

4. A health psychologist is modeling the probability that individuals engage in regular exercise (Yes = 1, No = 0) based on their age and daily hours of screen time. She fits a logistic regression, a type of generalized linear model (GLM) with a logit link.

Which of the following best interprets a negative coefficient for screen time in this model?

- A) As screen time increases, the chance of exercising decreases, holding age constant
- B) Screen time predicts lower actual exercise time in hours
- C) A negative coefficient means screen time has no effect on exercise likelihood
- D) The model assumes screen time is unrelated to the outcome

5. A researcher uses a linear regression model to predict anxiety scores based on daily screen time (in hours). After fitting the model, she finds that the coefficient of determination R^2 is 0.64. What is the best interpretation of this result?

A) 64% of the anxiety scores are predicted without any error
B) 64% of the **variation** in anxiety scores is explained by screen time
C) 64% of the participants had high anxiety due to screen time
D) Screen time causes anxiety in 64% of people

6. A researcher builds a multiple linear regression model to predict students' final exam scores based on two variables: number of hours studied and number of hours slept the night before the exam. The regression equation is:

$$\text{Score} = 30 + 4 * (\text{Hours Studied}) + 2 * (\text{Hours Slept})$$

Which of the following is the **best interpretation** of the coefficient for *Hours Studied*?

A) Students who study more always score higher than those who sleep more
B) Each additional hour studied increases the score by 4 points, regardless of sleep
C) For each additional hour studied, the expected test score increases by 4 points, holding hours slept constant
D) For each additional hour studied, the expected test score increases by 2 points, holding hours studied constant

7. A psychologist administers a standardized IQ test to a large group of participants. The scores are normally distributed with a mean of 100 and a standard deviation of 15. What percentage of participants are expected to have an IQ between 85 and 115?

A) 34%
B) 68%
C) 95%
D) Insufficient information

8. Which of the following is NOT a reason to use ANOVA instead of multiple t-tests?

A) Control for Type I error
B) Allows post-hoc comparisons
C) Higher chance of finding significance
D) Tests multiple means simultaneously

9. What is the purpose of Levene's Test in the context of ANOVA?

A) To test for normality across the data
B) To compare the variance in the data conditions
C) To analyze the effect size of the findings
D) To find outliers in the data

10. A pharmaceutical company recruited people to test their new drug, Y. Participants were randomly given a placebo or drug Y in the first 3 months. One year later, participants that had taken placebo earlier were now given the drug Y, and those who had taken the drug earlier were given a placebo. Which test is most appropriate to understand the effects of the drug?

- A) t-test
- B) Repeated Measures ANOVA
- ☒ C) Mixed ANOVA
- D) Paired t-test

11. A researcher is analyzing a questionnaire with 20 items measuring psychological well-being and is looking to reduce dimensionality. Which statement best differentiates Principal Component Analysis (PCA) from Factor Analysis (FA) in this context?

- A) PCA assumes items are caused by latent factors, while FA extracts uncorrelated components.
- ☒ B) PCA maximizes variance explained by components, while FA models underlying latent constructs.
- C) PCA is used for confirmatory analysis, while FA is used for exploratory analysis.
- D) PCA requires a larger sample size than FA to achieve reliable results.

12. A researcher conducts a between-subjects study to compare satisfaction levels (ordinal: low, medium, high) with a new therapy program between two groups (treatment vs. control, n=25 per group). Which nonparametric test is most appropriate for analyzing the difference in satisfaction levels?

- A) Wilcoxon Signed-Rank Test, because it compares paired ordinal data.
- ☒ B) Mann-Whitney U Test, because it compares independent groups on an ordinal outcome.
- C) Chi-Squared Test, because it analyzes continuous data between groups.
- D) Kruskal-Wallis Test, because it compares two independent groups.

13. A researcher is studying whether there is an association between gender (male, female) and preference for a new therapy program (yes, no) in a sample of 200 participants. Which of the following is the most appropriate answer for applying the Chi-Squared Test of Independence in this study?

- A) The data must be normally distributed.
- B) The sample size must exceed 300 participants.
- C) The variables must be measured on an interval scale.
- ☒ D) None of the above.

Descriptive Questions (32 marks):

Question 1 (6 marks)

A university wants to find out which of four study techniques helps students perform better in a competitive entrance exam. They assign 200 students randomly to one of four groups:

- Group A: Self-study
- Group B: Coaching classes
- Group C: Online courses
- Group D: Group study

After 3 months, all students take the same practice exam. The university wants to compare the average exam scores of the groups to see which techniques worked better. They perform six separate t-tests (e.g., A vs B, A vs C, B vs D, etc.) using a 5% significance level for each.

a) What is the problem with running six separate t-tests without adjusting the significance level? (1 mark)

Multiple comparison - could lead to a test giving significant results just because the other tests didn't as a balance for chance. This is a general problem with frequentist methods. Let's say all tests were supposed to give no significance $\rightarrow H_0$ should have been accepted. In a general case we would want $P(\text{false reject}) \leq 0.05$. However with 6 tests $\rightarrow P(\text{false reject}) = 1 - P(\text{accept all}) = 1 - (0.95)^6 \geq 0.05$. $\therefore$ The significance level should be adjusted accordingly. otherwise I am allowing for a larger margin of error.

b) Explain what it means to control the family-wise error rate and explain how it can be achieved through a Bonferroni correction. (2 marks)

FWER $\rightarrow$ The probability of getting even a single false positive result.

It aims to control the FP rate to be less than 0.05 (significance level).

	reject H_0	accept H_0
H_0 true	FP	
H_0 false		FN

Bonferroni correction:

Aims to control the FWER in cases of multiple comparisons.

If there are n tests, $\therefore p_1, p_2 \dots p_n$ are the p -values of the comparisons.

correction: multiple each p -value by n

or alternatively reduce significance level to α/n

Now compare the modified $p_i \times n$ values with α (or)

p -values with modified α/n

to determine significance.

How it helps:

$$P(\text{false reject}) = 1 - P(\text{all accept}) = 1 - \left(1 - \frac{0.05}{n}\right)^n \leq 0.05.$$

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c) What do False Discovery Rate (FDR) correction methods control? (1 mark)

FDR controls for the false discoveries with respect to all the discoveries.

FDR $\Rightarrow \frac{FP}{FP+TP} \rightarrow$ false discoveries
 $FP+TP \rightarrow$ all discoveries.

	accept H_0	reject H_0
H_0 false	FN	TP
H_0 true	TN	FP

correction $\rightarrow$ Benjamini-Hochberg.

similar to previous ex. say, $p_1, p_2 \dots p_n$.

change α from 0.05 to $\alpha \cdot \frac{i}{n}$

} sort in ascending

$\rightarrow$ then compare.

d) Mention one advantage and one disadvantage of each of the Bonferroni and Benjamini-Hochberg corrections. (2 marks)

Bonferroni $\rightarrow$ Advantage: strongly controls FP leading to no false discoveries.

Disadvantage: Too stringent. Could lead to missing out on significant results. (FN increases)

B-H

$\rightarrow$ Advantage: Not too stringent. Doesn't affect FN significantly.

Disadvantage: Does not control FP too strongly. Could lead to few false discoveries if p -values are not uniformly sorted.

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Disadvantage for both $\rightarrow$ They assume independent tests (comparisons) while in general cases they are not independent.

Question 2 (7 marks)

In a study comparing reaction times between two groups (a caffeine group and a placebo group), the data violate the assumptions of a t-test: the distributions are heavily skewed and contain outliers.

Describe how a permutation-based non-parametric test could be used to compare the mean reaction times between these two groups. Your answer should include:

1. What is the null hypothesis here? (1 mark)

Let μ_0 be the difference in means b/w two groups on the original data.

null hypothesis: $H_0 \rightarrow$ There is no difference in the reaction times between the two groups.

$$\mu_0 = 0.$$

$\Rightarrow$ That the current group is as equal to a null distribution created randomly with the same values.

2. The key steps involved in performing the test are: (5 marks)

$\rightarrow$ compute the $\mu_0 \rightarrow$ original difference b/w means. (say caffeine group - placebo)

$\rightarrow$ Repeat for 10000 iterations

- merge both the groups and randomly split in the original distribution ratio $\rightarrow$ null distribution.

- compute μ as difference of mean in the two groups just created.

- if $\mu_0 > \mu$ then increase some variable count by 1.

$\rightarrow$ Compute the p-value as $\frac{\text{count}}{10000} \rightarrow$ ~~the~~ number of iterations

$\rightarrow$ if $p \leq 0.05$ (or other significance level)
then reject the null distribution $\therefore$ significant result.

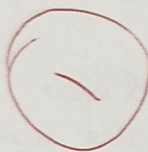
else

accept null $\rightarrow$ no significant result.

$\rightarrow$ for visualization, plot the bootstrap distribution of μ computed in every iteration and check that μ_0 lies outside the confidence interval of the bootstrap distribution.

4

3. How is the p-value calculated from the permutation "null" distribution? (1 mark)


$$p\text{ value} = \frac{\text{no of null dist where } \mu_0 > \mu}{\text{total num of iterations.}}$$

It represents how likely is the given split of groups.
if $p < 0.05$ then there is significant difference b/w groups.

Question 3 (8 marks)

You are consulting for a health psychologist who has collected the following data from 300 adult participants as part of a study on stress and lifestyle:

- Perceived stress score (0–40)
- Daily caffeine intake (mg)
- Physical activity group (Sedentary, Moderate, High)
- Whether the person reports frequent insomnia (Yes/No)
- Number of sick days taken in the past year

They ask you the following questions:

- a) Does caffeine intake differ across the three physical activity groups?
- b) Is there an association between insomnia and physical activity level?

c) Can stress scores be predicted from caffeine intake?

d) What is the best way to model the number of sick days as a function of stress score and insomnia status?

all tests are parametric so I am assuming that normality is always satisfied.
For each question (a-d), identify the most appropriate statistical method (from t-test, ANOVA, chi-square test, regression, or GLM) and justify your choice based on the types of variables (continuous, categorical, count, etc) and the research question. 1 marks for clearly identifying the variable types (both IV and DV) and 1 marks for picking the right method with a justification based on the research question. Total = $4 \times 2 = 8$ marks.

assumption: all data points are independent.

a)

Types of variables:

Physical activity group $\rightarrow$ ~~IV~~ ^{IV} $\rightarrow$ categorical.
Caffeine intake $\rightarrow$ DV $\rightarrow$ numerical continuous

Method and justification:

categorical $\rightarrow$ numerical. need to test for comparison difference between distributions means in the 3 groups.
multiple IV categories (> 2) and single DV $\rightarrow$ ANOVA is the best test.
assumption: variance is homogeneous. otherwise we'd use ANCOVA.

b)

Types of variables:

Physical activity group $\rightarrow$ IV $\rightarrow$ categorical.
Insomnia $\rightarrow$ DV $\rightarrow$ ~~count~~ (convert boolean to count) categorical per group.

Method and justification:

categorical $\rightarrow$ count. need to compare if the number/frequency of insomnia is consistent across the 3 groups $\rightarrow$ Chi-Square test.
It is also non parametric so does not depend on the distribution of data.
E.g. ~~the~~ in the ratio of the original distribution in each group
The aim is to see if the the insomnia dist matched the E.g. dist.

c)

Types of variables:

Caffeine intake $\rightarrow$ IV $\rightarrow$ numerical continuous.
Stress score $\rightarrow$ DV $\rightarrow$ numerical ~~continuous~~ count (but can be modelled as continuous)

Method and justification:

predictors and outcomes are both numerical so simple regression can be used.
assuming normality of distribution of residuals in the data.

d)

Types of variables:

stress score $\rightarrow$ IV $\rightarrow$ numerical, ~~continuous~~ count.
categorical = insomnia status $\rightarrow$ DV $\rightarrow$ can be converted to 1/0 numerical.
sick days $\rightarrow$ DV $\rightarrow$ numerical (count)

Method and justification:

since predictors and outcomes are both numerical and there are 2 IV, some form of multiple regression can be used.

This is assuming normality of residuals is satisfied.

A better case: since sick days is a count variable, and residuals need not be normal, a GLM with poisson regression and log link can be used.

— Ans.

Question 4 (4 marks)

Give examples to explain when to choose a one-tailed versus a two-tailed hypothesis test. Provide scenarios/examples where one is more appropriate than the other.

A one-tailed test only checks for significance along an estimated direction (higher/lower) whereas a two-tailed test tests for significance across both directions.

when to choose one-tailed test:

- Belief or strong evidence that the significance is along one direction only
- The test statistic is more interpretable for a one-tailed test rather than a two-tailed test. Eg. if the distribution is an F distribution, the lower tail doesn't mean much. we can just take the higher tail.

when to choose two sided test

- no prior assumption of the directionality.
- test statistic is significant on both sides. Both lower and upper tail provide meaningful interpretation. eg if distribution is a t-dist ideally / generally we ~~can~~ should do a two sided test.

Examples:

- Effect of a drug on exam scores where two independent groups are given a drug and other a placebo. Does drug affect exam score? two-tailed test as we do not know if it would lead to better / worse scores.
- Does exam preparation ~~effect~~ ~~exam~~ give more ~~score~~ than no preparation. One-sided test as there is a prior belief that it does.
- In cases like regression / ANOVA, the original ~~of~~ hypothesis is tested with a one sided F-test. However the effect of regression coeffs and/or ANOVA comparisons (for post hoc) are generally done as a two sided tests.

Question 5 (4 marks)

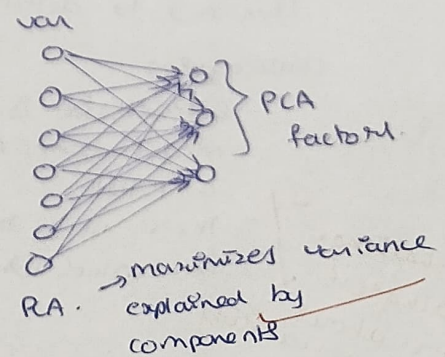
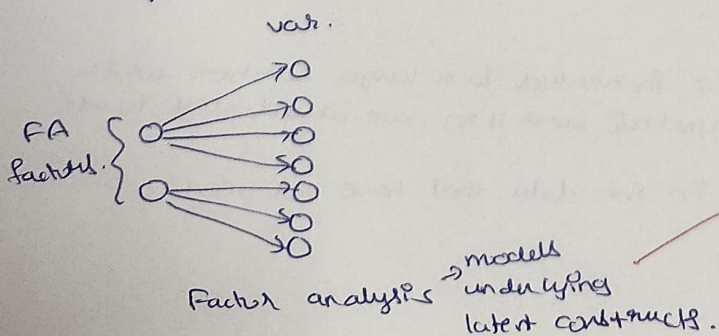
What is multicollinearity, and explain how it can impact the interpretation of a multiple regression model in a behavioral study? Given that Principal Component Analysis (PCA) and Factor Analysis (FA) both help address multicollinearity, what are the essential differences between them? Illustrate with an example to bring out the differentiation.

Multicollinearity $\rightarrow$ when multiple predictors are correlated to each other. Highly correlated predictors could lead to a lot of problems in statistical analysis. Variance Inflation factor is one way to determine if the variables present are correlated ($VIF > 5$)

problems with multicollinearity $\rightarrow$

- $\rightarrow$ curse of dimensionality $\rightarrow$ model may overfit to the noise as there are redundant dimensions which don't contribute to ~~any~~ extra significant information.
- $\rightarrow$ Regression coefficients would be worse than just having a single predictor. AIC would be higher for the model with multicollinearity.

ways to remove multicollinearity $\rightarrow$ FA and PCA.



→ factor analysis

- Tries to encounter ~~hidden~~ variables that gave rise to the data values.
- It is more suggestive than ~~explorative~~.
- Multiple factors can be made with varying weights. There is no fixed factors which are the best. The factors may be non independent.
- Eg: Questionnaire. The problem setter will know what hidden variables need to be measured and assign weights from these hidden variables to the actual questions. *mention it*

→ Principle component analysis.

- Tries to project the data to new axes which best explain the maximum variance. Each axes correspond to independent high variance axes.
- It is ~~explorative~~, used to find optimal no of variables ~~so~~ that can capture most of the data.
- Fixed ~~as variable~~ factors corresponding to the principle components.
- Eg: multiple personality traits values are ~~many~~ many and we want to capture the most significant axes which will explain the data with ~~less~~ number of variables.

Question 6 (3 marks)

A researcher conducts a between-subjects study comparing stress levels (1=very low to 5=very high) between two groups (mindfulness intervention vs. control, $n=25$ per group). The data and residuals are not normally distributed. What statistical test would you suggest and why? Also, list potential challenges you might encounter when applying your proposed test.

Categorical predictors with numerical outcome.

The two groups are independent. we care about the difference rather than correlation.

Since data and residuals are not normal → non-parametric test.

Based on the scenario the ideal test would be Mann-Whitney U-Test. ✓

since we want a non parametric alternative to independent-samples t-test.

The outcome is also ordinal making it easy for determining ranks in the test. The test involves computing the U statistic for each group, taking the minimum and ensuring it is below the standard limit (for n) to determine significant difference.

Challenges →

- Dataset size is small. Could be increased to a larger number which might even make it parametric and then we would need to use a t-test.
- There may be confounds in the data that have not been accounted for. *what?*

2.5
not challenge during test
it's about reliability